

Simulation Evidence Qualification Module (SEQM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS)

Module: SEQM — Simulation Evidence Qualification Module

Operational Layer: Simulation Evidence Qualification Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Core Module

Governed Space: Simulation Evidence Qualification

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AI-Readable: Yes

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Canonical Language: English

Canonical Definition

Simulation Evidence Qualification Module (SEQM) is the first Core Module of the Simulation Evidence & Reliance Boundary Standard (SERBS) within SIMULOS® — Simulation Governance Architecture.

SEQM governs the transformation of simulation-derived outputs, observations, correlations, comparisons, and governed inferences into explicitly qualified Simulation Evidence for a defined evidential purpose.

The Module establishes whether a simulation-derived information object is sufficiently identified, traceable, context-bound, purpose-linked, condition-aware, and methodologically interpretable to be treated as governed evidence within the SIMULOS® lifecycle.

SEQM does not determine the ultimate strength, sufficiency, or validation consequence of that evidence.

Its task is narrower and foundational:

to determine whether simulation-derived information legitimately qualifies as evidence at all, and to define the evidential basis under which that qualification is valid.

Core Question

What simulation-derived information legitimately qualifies as evidence, and for what evidential purpose?

Architectural Position

SEQM is the first module within SERBS.

It receives governed information from:

Simulation Output Governance Standard (SOGS)

and

Simulation-to-Reality Correlation Standard (SRCS).

It precedes:

SEWM — Simulation Evidence Weight Module.

Its architectural role is therefore to establish the evidence object before any attempt is made to assign evidential weight, define reliance boundaries, impose use limitations, or transfer the evidence into broader validation governance.

Internal Flow Position

Governed Simulation Output Governed Inference Basis Simulation-to-Reality Correlation Basis → Evidence Candidate Identification → Evidential Purpose Definition → Simulation Basis Traceability → Qualification Condition Assessment → Applicability Definition → Uncertainty Preservation → Evidence Qualification State → Qualified Simulation Evidence → SEWM — Simulation Evidence Weight Module

Operational Space

SEQM governs the point at which simulation-derived information crosses from:

simulation result

into:

qualified simulation evidence.

This operational space includes: evidence candidate identification; evidential purpose definition; proposition linkage; simulation basis traceability; output provenance; inference provenance; correlation basis linkage; contextual qualification; applicability definition; qualification conditions; uncertainty preservation; temporal applicability; version applicability; scenario applicability; configuration applicability; evidence qualification state; evidence disqualification where the required basis is absent.

Operational Scope

SEQM applies to any simulation-derived information that an organization intends to use as evidence.

Such information may include: numerical outputs; state transitions; observed simulated behavior; event frequencies; failure patterns; scenario outcomes; threshold crossings; sensitivity results; comparative results; stochastic distributions; trajectory results; predicted behaviors; simulated interactions; failure modes; anomaly patterns; scenario coverage observations; simulation-to-reality comparisons; governed simulation inferences.

The Module applies equally to: a single simulation execution; repeated runs; scenario portfolios; Monte Carlo campaigns; high-performance simulation campaigns; digital twins; physical simulations; hybrid simulations; mission simulations; engineering simulations; scientific simulations; safety simulations; AI simulations; autonomous-system simulations; future simulation technologies.

Evidence Candidate Principle

A simulation-derived result must first be identified as an Evidence Candidate before it can be qualified.

An Evidence Candidate is:

a defined simulation-derived information object proposed for use in supporting, challenging, comparing, informing, or constraining a specific proposition, question, assessment, or decision context.

An Evidence Candidate may originate from: one execution; multiple executions; one scenario; a scenario set; a simulation campaign; a correlation result; a pattern across outputs; a governed inference.

But existence alone does not establish evidential status. Simulation Output $\neq$ Simulation Evidence

SEQM enforces the distinction:

Simulation Output $\neq$ Simulation Evidence

A result becomes governed evidence only when the relationship between that result and an explicit evidential purpose is established.

For example:

"The simulator produced a maximum temperature of 912°C."

is an output statement.

"Under Scenario S-47, Configuration C-3, and the governed assumptions of Simulation Model M-12, the simulation produced a maximum temperature of 912°C, supporting evidence concerning thermal behavior under those represented conditions."

is an evidentially qualified statement.

The difference is governance context.

Evidential Purpose Principle

Every qualified Simulation Evidence object must have a defined Evidential Purpose.

The Evidential Purpose specifies what the evidence is intended to inform.

It may relate to: a hypothesis; a design question; a performance question; a safety proposition; a comparative assessment; a mission condition; a failure hypothesis; a scenario question; a behavioral proposition; a threshold condition; an engineering claim; a validation input; an operational question.

Evidence without a defined evidential purpose remains analytically informative but methodologically underqualified.

Evidence-to-Proposition Relationship

SEQM requires a defined relationship between:

Simulation Evidence and the proposition or question the evidence is intended to inform.

The relationship may be: supporting; challenging; contradicting; comparative; conditional; exclusionary; sensitivity-based; exploratory.

This distinction matters because the same simulation result may legitimately serve different evidential functions in different contexts.

A failure result may weaken one proposition while strongly supporting another.

Simulation Basis Traceability

Qualified Simulation Evidence must remain traceable, where materially relevant, to the governed simulation basis from which it originated.

The traceability chain may include:

Simulation Evidence → Simulation Output → Execution Record → Scenario → Input Basis → Model & Environment → Assumptions → Reality Representation → Purpose & Scope → Simulation Object

Where SRCS applies:

Simulation Evidence →

Simulation-to-Reality Correlation Basis

This chain ensures that evidential meaning cannot become detached from the conditions that created it.

Evidence Context Principle

Simulation evidence is inherently context-dependent.

SEQM therefore requires relevant evidential context to remain visible.

Context may include: Simulation Object identity; object version; configuration; purpose; scope; excluded conditions; reality representation; abstraction level; fidelity level; assumptions; model version; environment version; input set; temporal basis; scenario; parameter range; execution conditions; output type; correlation basis; uncertainty.

Context is not metadata decoration.

It is part of the evidential meaning.

Qualification Conditions

An Evidence Candidate may be qualified only where the minimum material conditions necessary to interpret it are established.

Qualification conditions may require: identifiable evidence object; identifiable originating output; defined evidential purpose; traceable simulation basis; known Simulation Object; known object version or configuration; known scenario basis; known execution source; known material assumptions; known output meaning; known correlation basis where reality claims are involved; known uncertainty affecting interpretation; known applicability conditions; known material exclusions.

The precise combination depends on the evidential purpose.

SEQM therefore does not impose unnecessary uniformity.

It requires sufficient contextual definition for legitimate evidential interpretation.

Evidence Qualification State

SEQM establishes explicit qualification states.

A Simulation Evidence Candidate may be classified as: 1. Unqualified

The simulation-derived information exists but lacks sufficient evidential context or purpose definition. 2. Conditionally Qualified

The information may function as evidence only under explicitly defined conditions or limitations. 3. Qualified

The information has sufficient simulation-specific basis, traceability, purpose alignment, and contextual definition to function as governed Simulation Evidence. 4. Qualification Suspended

A previously qualified evidence object has encountered a material change, unresolved discrepancy, missing provenance, or other condition requiring reassessment. 5. Disqualified

The information cannot legitimately function as Simulation Evidence for the stated evidential purpose.

Qualification state must remain distinguishable from evidential weight.

Qualified ≠ strong evidence.

Qualification determines whether the object may enter evidence governance.

SEWM determines how much simulation-specific evidential weight it may carry.

Conditional Qualification

Many simulation results can be evidentially valid only under specified conditions.

SEQM therefore permits conditional qualification.

Example:

Qualified only for Object Version V4, environmental range E2–E5, Scenario Family S3, and model configuration M7.

Such evidence remains legitimate.

Its qualification is simply bounded.

Conditional qualification must not be silently transformed into universal qualification downstream.

Negative Evidence Qualification

SEQM explicitly allows negative, failed, anomalous, unstable, or contradictory results to qualify as evidence.

Evidence qualification must not depend upon whether the result supports the preferred hypothesis.

Examples include: failure to achieve mission objectives; model divergence; repeated instability; scenario-specific failure; unexpected collision; thermal exceedance; non-convergence; behavioral anomaly; disagreement with operational data; unexpected sensitivity.

If the result is relevant, traceable, and contextually interpretable, it may qualify as evidence.

Inference Qualification

Some simulation evidence is directly observed in simulation outputs.

Other evidence is derived through inference.

SEQM requires this distinction to remain visible.

Direct Simulation Observation ≠ Derived Simulation Inference

Where evidence depends upon an inference, the qualification basis must identify: originating outputs; inference method; inferential assumptions; transformation or aggregation; inference scope; uncertainty introduced by

inference.

The existence of a governed inference does not automatically increase or decrease evidence weight.

It simply changes the evidential pathway that must remain traceable.

Aggregated Evidence Qualification

Simulation campaigns frequently aggregate outputs across large numbers of runs.

SEQM requires aggregated evidence to identify: aggregation basis; included executions; excluded executions; scenario composition; common assumptions; common model dependencies; statistical method; aggregation transformation; treatment of failed runs; treatment of anomalous runs; dependence structure where known.

An aggregate metric must not obscure the evidence objects from which it was created.

Evidence Independence Awareness

SEQM does not determine final evidential weight, but it must preserve information relevant to later assessment of evidence dependence.

Examples include: repeated use of the same model; shared assumptions; identical input generation; common scenario design; shared execution infrastructure; common data transformations; common correlation basis.

This enables SEWM to distinguish between:

many evidence objects

and

many independently informative evidence objects.

Applicability Definition

Every qualified evidence object should identify the context within which its evidential meaning applies.

Applicability may be defined by: object; configuration; scenario; environment; parameter range; temporal interval; operational condition; mission phase; behavior class; model state; input family; uncertainty range.

Applicability defines where the evidence may meaningfully enter subsequent governance.

It does not yet define the maximum reliance boundary; that is owned by SRBM.

Temporal Qualification

Evidence qualification may depend upon time.

Evidence can become underqualified where: the Simulation Object changes; a model is replaced; assumptions become obsolete; new operational data appears; reality correlation changes; input distributions materially shift; scenario relevance changes.

SEQM must therefore support:

Qualification Review and Qualification Reassessment.

This creates the interface to SLRS lifecycle governance.

Evidence Provenance

SEQM requires simulation evidence to preserve sufficient provenance to identify how the evidence came into existence.

Provenance may include: originating simulation; originating execution ID; simulation campaign ID; scenario ID; model version; environment version; object version; input package; output reference; transformation history; inference history; aggregation history; qualification record.

SEQM governs the simulation-specific evidential provenance relationship.

General provenance integrity and audit assurance remain outside its boundary where governed by INTEGROS® or ADIT®.

Qualification Failure Conditions

An Evidence Candidate should not qualify where material information necessary to understand its evidential meaning is absent.

Examples include: unknown originating execution; unknown Simulation Object version; missing scenario basis; unclear evidential purpose; untraceable transformed output; unidentified material assumption; unknown model configuration; missing relationship to the proposition being supported; output used outside its represented context; correlation claim without a governed correlation basis; aggregation that conceals materially relevant failures; evidence whose meaning cannot be reconstructed.

Qualification failure does not necessarily mean the underlying simulation is invalid.

It means the information cannot currently function as governed Simulation Evidence for the intended purpose.

SEQM Boundary with SOGS

SOGS governs: what the simulation produced; output identity; output classification; inference; output limitation; output documentation.

SEQM governs: whether that governed output or inference may legitimately function as evidence; for what evidential purpose; under what qualification conditions; with what simulation basis.

Therefore:

Governed Output ≠ Qualified Evidence.

SEQM Boundary with SRCS

SRCS governs the correlation relationship between simulation and reality.

SEQM may use that governed correlation relationship as part of an evidence qualification basis.

It does not recalculate or redefine correlation.

Therefore:

Correlation Result → Evidence Qualification Input

not

Evidence Qualification → Correlation Determination.

SEQM Boundary with SEWM

SEQM determines whether simulation-derived information qualifies as evidence.

SEWM determines the simulation-specific evidential weight of that qualified evidence.

Therefore:

Qualification ≠ Weight.

A result may be fully qualified yet carry weak evidential weight.

A result may also be conditionally qualified yet carry strong evidential weight within a narrow context.

SEQM Boundary with VALIDOS®

SEQM does not determine whether evidence is sufficient for a broader Validation Determination.

It governs only the qualification of simulation-derived information as Simulation Evidence.

The boundary is:

SEQM: Does this simulation-derived information legitimately qualify as evidence for this defined evidential purpose?

VALIDOS®: Does the total validation evidence basis sufficiently support the Validation Determination and intended reliance?

SEQM Boundary with INTEGROS®

SEQM does not own general evidence integrity.

It may require evidence provenance, traceability, and contextual completeness as conditions of qualification.

INTEGROS® governs broader integrity properties of the evidence and its supporting structures.

SEQM Boundary with ADIT®

SEQM may require a reconstructable evidential basis.

ADIT® may audit whether records, evidence, observations, and execution traces are sufficiently preserved and verifiable.

SEQM does not own auditability governance.

Minimum Implementation Framework

A minimum SEQM implementation should establish: 1. Evidence Candidate Identification

Identify the exact simulation-derived information proposed as evidence. 2. Evidential Purpose Definition

Define the proposition, question, comparison, or decision context the evidence is intended to inform. 3. Evidence Source Identification

Identify the originating outputs, observations, correlations, comparisons, or inferences. 4. Simulation Basis Traceability

Link the Evidence Candidate to the relevant governed simulation lifecycle elements. 5. Context Definition

Identify object, version, configuration, scenario, model, environment, assumptions, inputs, and other materially relevant conditions. 6. Applicability Definition

Define where the evidential meaning applies. 7. Uncertainty Preservation

Preserve material uncertainty affecting interpretation. 8. Qualification Condition Assessment

Determine whether the minimum required conditions for legitimate evidential interpretation are satisfied. 9. Qualification State Assignment

Assign the Evidence Candidate a governed qualification state. 10. Qualification Record Creation

Document the basis supporting the qualification decision.

Governance Outputs

SEQM may produce: Simulation Evidence Candidate Record Simulation Evidential Purpose Statement Simulation Evidence Source Record Simulation Evidence Traceability Chain Simulation Evidence Context Record Simulation Evidence Applicability Statement Simulation Evidence Qualification Condition Record Simulation Evidence Uncertainty Link Simulation Evidence Qualification State Simulation Evidence Disqualification Record Simulation Evidence Qualification Record Qualified Simulation Evidence Object

These outputs provide the input basis for SEWM — Simulation Evidence Weight Module.

Enterprise Application

SEQM enables organizations to prevent a common simulation governance failure:

treating every technically generated result as evidence.

An organization may use SEQM to determine: whether a particular simulation result legitimately supports a stated proposition; whether the result remains connected to the correct Simulation Object version; whether scenario and execution context are preserved; whether a derived inference is traceable to its source outputs; whether failed and anomalous outcomes remain evidentially visible; whether aggregated campaign results conceal material dependencies; whether evidence is still applicable after material change; whether an output should remain unqualified, become conditionally qualified, become fully qualified, or be disqualified; whether simulation-derived information is ready to proceed into evidential weight assessment.

This becomes especially important in very large simulation programs where millions or billions of outputs may exist but only a subset may legitimately function as evidence for a particular claim.

SEQM therefore creates an explicit methodological gate between simulation production and simulation evidence governance.

Foundational Principles

Simulation output is not automatically simulation evidence.

Evidence qualification requires an explicit evidential purpose.

Evidence must remain connected to the simulation basis that gives it meaning.

Qualification is contextual, not universal.

Qualified evidence may still carry weak evidential weight.

Conditional evidence must remain conditional.

Negative simulation results may qualify as evidence.

Aggregated evidence must not erase its originating simulation basis.

Evidence provenance is part of evidential qualification.

A simulation result that cannot be meaningfully traced, contextualized, and interpreted should not be treated as governed Simulation Evidence.

Canonical Closing Statement

Simulation Evidence Qualification Module establishes the first evidential control point within the Simulation Evidence & Reliance Boundary Standard by transforming simulation-derived outputs, observations, correlations, comparisons, and governed inferences into explicitly identified, purpose-linked, traceable, context-bound, applicability-defined, uncertainty-aware, and qualification-state-controlled Simulation Evidence. Through governance of Evidence Candidates, evidential purpose, proposition relationships, simulation basis traceability, contextual conditions, provenance, inference pathways, aggregation, applicability, qualification states, temporal relevance, and qualification failure, SEQM prevents simulation results from acquiring evidential status merely because they were computationally produced or analytically persuasive. By separating evidence qualification from evidential weight, reliance, validation sufficiency, integrity, and auditability, SEQM preserves the architectural boundaries of SIMULOS®, VALIDOS®, INTEGROS®, and ADIT® while creating the governed evidence object required for downstream evidence assessment. No simulation-derived result should legitimately enter broader evidential reliance until the conditions under which it becomes evidence have first been made explicit.

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